Hopefully these last chapters have successfully conveyed the way in which Many-Worlds quantum mechanics might have significant implications for the long-standing problem of quantum gravity. To be honest, many physicists working on these problems don't think of themselves as using Many-Worlds, though they are implicitly doing so. They certainly are not using hidden variables, or dynamical collapses, or an epistemic approach to quantum mechanics. When it comes to understanding how to quantize the universe itself, Many-Worlds seems to be the most direct path to take, if nothing else.

Is the picture we've sketched, where the entanglement between degrees of freedom somehow comes together to define the geometry of our approximately classical spacetime, actually on the right track? Nobody knows for sure. What seems clear, given the current state of our knowledge, is that both space and time could emerge from an abstract quantum state in the desired way—all the ingredients are there, and it's not out of place to hope that a few more years of work will bring a much sharper picture into focus. If we train ourselves to discard our classical prejudices, and take the lessons of quantum mechanics at face value, we may eventually learn how to extract our universe from the wave function.

---

* It's not completely agreed upon that infalling objects actually do travel deep into the interior of a black hole. In 2012 a group of physicists argued that, if information is going to escape from evaporating black holes without violating the basic tenets of quantum mechanics, something dramatic has to happen at the event horizon: not quiet, empty spacetime, as is usually assumed, but a blast of high-energy particles known as a firewall. Opinions about the firewall proposal are divided, as theorists continue to argue back and forth about the issue.